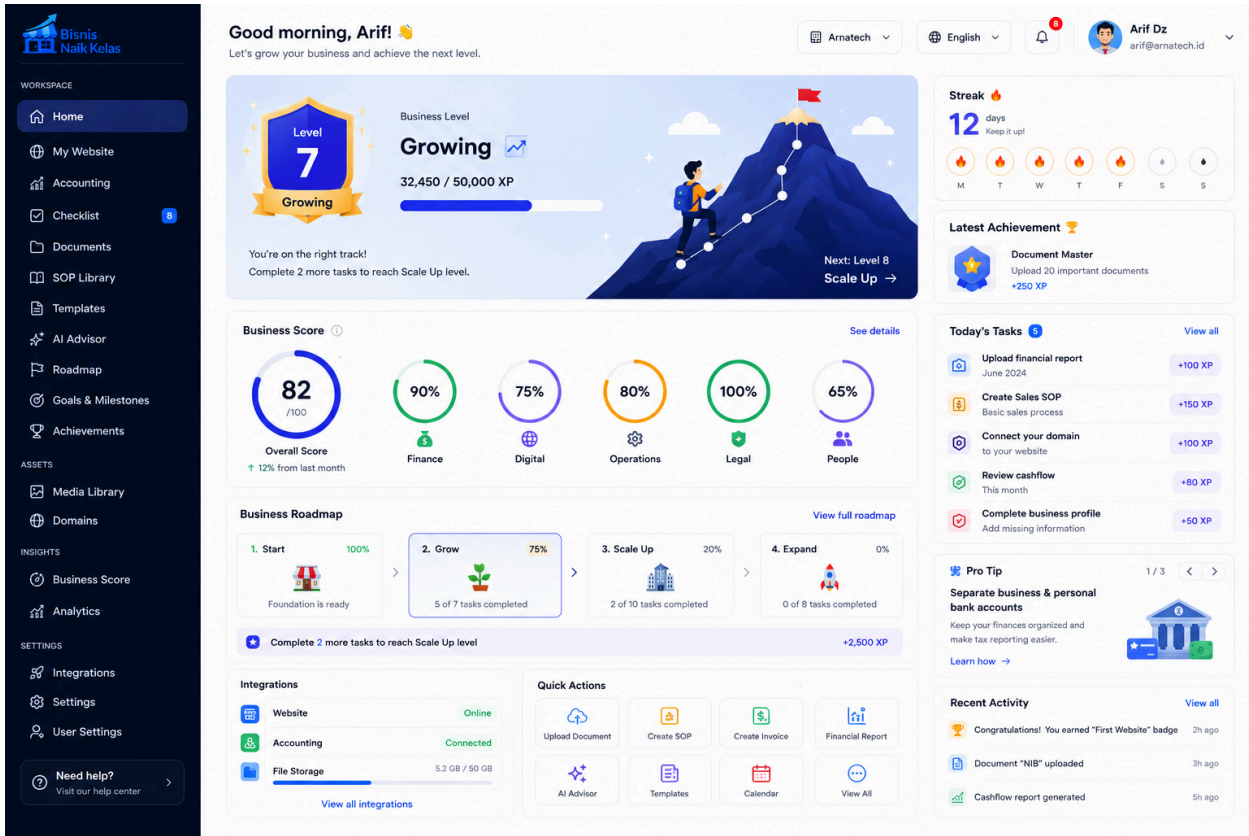


PRD - Business Hub Service

Bisnis Naik Kelas master product requirements document for backend, frontend, QA, DevOps, and AI integration

Field	Value
Product name	Business Hub
Service name	arna_business_hub
Repository name	arna_business_hub
Docker image	arnatechid/arna-business-hub:<build-number>
Swarm service	arna-business-hub
Database	arna_business_hub_db
Document version	1.0
Date	2026-07-06
Owner	Arif Dzikrullah / Arnatech
Primary audience	Software Engineer, Frontend Engineer, QA Engineer, DevOps, Product



1. Document Overview

Business Hub is the central orchestration service for the Bisnis Naik Kelas ecosystem. It provides guided business growth through roadmap, checklist, business profile, vault metadata,

SOP and template workflows, gamification, readiness scoring, AI recommendations, and cross-service progress aggregation.

This PRD follows the structure and engineering depth of the Arna SME Accounting Service PRD. It is intended to reduce implementation ambiguity across backend, frontend, QA, DevOps, and product delivery.

1.1 Product Context

Bisnis Naik Kelas is not positioned as a traditional SaaS collection of unrelated tools. It is a guided business operating system for SMEs that helps users move from selling informally into a more organized, measurable, professional, scalable, and bankable business.

Business Hub is the product layer that makes this promise visible. Website Service builds identity. SME Accounting tracks financial health. File Manager stores documents. SSO manages users and organizations. Commerce manages packages and entitlements. Business Hub connects these services into a step-by-step journey.

1.2 Service Responsibility

- Own business journey, roadmap, checklist, level, XP, achievement, score, goals, calendar, vault metadata, SOP/template metadata, AI insight metadata, integration status aggregation, and dashboard orchestration.
- Consume status and facts from Website Service, SME Accounting, File Manager, Commerce, SSO, Notification, and AI Gateway.
- Never duplicate login, organization, storage, website publishing, accounting records, payment, or AI inference responsibilities already owned by other services.

2. Background

Indonesian SMEs often have sales activity but lack business structure. The common problems are informal identity, scattered records, mixed personal and business funds, manual operations, weak documentation, low compliance readiness, and unclear growth strategy.

The attached Bisnis Naik Kelas strategy materials define the core insight clearly: SMEs need guidance, not more disconnected tools. They seek outcomes such as better trust, clearer profit, repeat customers, and readiness to grow, not feature-heavy dashboards.

2.1 Core Problems to Solve

- Lack of professional identity and consistent business profile.
- Disorganized financial management and low cashflow visibility.
- Absence of operational systems, SOP, task ownership, and documentation.
- No centralized business document memory.
- No step-by-step guidance for what to improve next.

- No readiness framework for funding, compliance, partnership, or scale.
- Too many tools causing low adoption and confusion.

2.2 Product Philosophy

The product philosophy is: Free helps SMEs run their business; Premium helps SMEs grow and scale. Business Hub must preserve this philosophy through a generous free journey, then unlock more automation, benchmarking, AI, advanced templates, and deeper readiness features in Premium.

3. Goals

3.1 Business Goals

1. Increase perceived value of Bisnis Naik Kelas beyond website and accounting features.
2. Turn the platform into a daily business operating habit for SME owners.
3. Encourage Free-to-Premium upgrade through meaningful progress, automation, and growth readiness.
4. Create a structured business maturity data layer that can support future financing, compliance, and advisory offerings.
5. Make Business Hub the flagship workspace that unifies Website, Accounting, Documents, AI, and Notifications into one user journey.

3.2 Product Goals

6. Give every SME a clear step-by-step roadmap from informal selling to scalable business.
7. Provide a gamified dashboard with level, XP, streak, quest, score, and next actions.
8. Connect checklist progress to real service events such as website published, NIB uploaded, cashflow available, or SOP created.
9. Store business document metadata and relationships while using File Manager for actual files.
10. Provide Business Score across financial, digital, legal, operational, HR, sales, and customer dimensions.
11. Generate contextual AI recommendations based on roadmap, checklist, score, documents, website, and accounting facts.
12. Support Free and Premium package behavior through Commerce entitlement checks.

3.3 Engineering Goals

13. Build an independent backend service following Arna Service Boilerplate patterns.
14. Use Arna SSO for authentication, user identity, organization, roles, and permissions.
15. Use Arna Commerce for package, subscription, entitlement, and quota enforcement.
16. Use Arna File Manager for uploads/downloads and store only file references.

17. Use Website Service and SME Accounting through APIs/events, never by direct database access.
18. Use PostgreSQL as the main datastore and Redis for cache/background coordination when needed.
19. Use HTTP/JSON APIs for dashboard operations and Pulsar for async cross-service events where useful.
20. Provide OpenAPI documentation, Docker, Jenkins, health checks, logs, metrics, tests, and deployment readiness.

4. Non-Goals

21. Business Hub will not create its own authentication, organization, role, or subscription system.
22. Business Hub will not upload files directly to S3/object storage or issue permanent presigned URLs.
23. Business Hub will not replace Website Service, website builder, domain, SSL, or site publishing functions.
24. Business Hub will not create or mutate accounting transactions directly; it consumes summarized facts or events from SME Accounting.
25. Business Hub will not become a full project management system in MVP.
26. Business Hub will not become Google Drive, Notion, or a complex document editor.
27. Business Hub will not provide legal, tax, audit, or financing guarantee. It provides readiness guidance only.
28. Business Hub will not expose sensitive cross-tenant benchmarking or revenue data.
29. Business Hub will not require Enterprise-only customization in MVP.

5. Target Users

User	Need	Primary Screens
SME Owner	Know what to do next, see business progress, understand score, unlock growth steps.	Home, Roadmap, Checklist, Score, AI Advisor
Business Admin	Upload documents, complete checklist, maintain SOP/template documents, manage calendar tasks.	Checklist, Vault, SOP Library, Templates, Calendar
Team Member	Follow assigned tasks and see relevant business documents based on permission.	Checklist, Vault, SOP
Platform Admin	Troubleshoot tenants, entitlement, integrations, score jobs, and service health.	Admin API, logs, integration status
Engineer / QA	Implement and validate service contracts, tenant isolation, and UI states.	OpenAPI, tests, fixtures

6. Package Scope and Positioning

Business Hub follows Bisnis Naik Kelas package positioning: Free, Premium, and future Enterprise. MVP must support Free and Premium. Enterprise behavior may be defined as future extension but not required for MVP implementation.

6.1 Free Package

- Basic Business Profile
- Basic Roadmap and Checklist
- Basic Business Score
- Limited Vault metadata records
- Basic SOP/template browsing
- Daily/weekly quest basics
- Manual progress completion where integration is unavailable
- Basic AI tips if global AI quota allows

6.2 Premium Package

- Advanced roadmap by industry/business level
- More checklist automation and richer integrations
- Advanced Business Score breakdown and historical trend
- Unlimited or higher vault metadata quota
- SOP and template clone workflows
- Advanced quest, achievement, and milestone tracking
- AI Business Advisor with deeper cross-service context
- Calendar reminders and Notification integration
- Export readiness report

6.3 Enterprise Future

- Custom business roadmap and custom maturity model
- Multi-branch organization support
- Advanced RBAC and approval workflow
- Custom integration adapters
- Dedicated onboarding and consulting workflow
- Benchmarking and executive report pack

7. Package Entitlements

Feature access must never be inferred from package name directly. Business Hub must request entitlement keys from Arna Commerce and enforce them server-side.

Entitlement	Value Type	Purpose
-------------	------------	---------

business_hub.enabled	true/false	Gate access to Business Hub
business_hub.plan	free/premium/enterprise	Display and analytics only; do not use alone for access
business_hub.max_vault_items	integer/unlimited	Number of BusinessDocument metadata records
business_hub.max_custom_goals	integer/unlimited	User-created goals
business_hub.sop_library_enabled	true/false	SOP browse access
business_hub.sop_clone_enabled	true/false	Clone SOP to Vault
business_hub.template_center_enabled	true/false	Template browse access
business_hub.template_clone_enabled	true/false	Clone template to Vault
business_hub.ai_advisor_enabled	true/false	AI Advisor access
business_hub.ai_monthly_insights	integer/unlimited	AI recommendation quota
business_hub.advanced_score_enabled	true/false	Advanced score dimensions
business_hub.score_history_enabled	true/false	Historical score trend
business_hub.calendar_reminder_enabled	true/false	Business Calendar reminders
business_hub.gamification_enabled	true/false	XP, level, quest, achievements
business_hub.leaderboard_enabled	true/false	Optional leaderboard
business_hub.readiness_export_enabled	true/false	Export readiness report

8. Feature Matrix

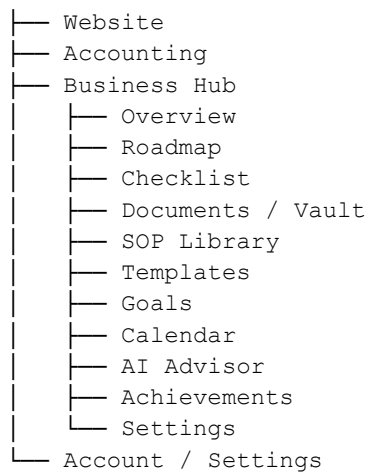
Feature	Free	Premium	Enterprise Future
Home gamified dashboard	Yes	Yes	Customizable
Business profile	Basic	Advanced fields	Custom fields
Roadmap	Default	Industry-aware	Custom
Checklist	Default	Dynamic + auto complete	Custom workflow
Business Vault metadata	Limited	Higher/unlimited	Advanced taxonomy
File storage	Via File Manager	Via File Manager	Via File Manager
SOP library	Browse	Browse + clone	Custom library
Template center	Browse	Browse + clone	Custom templates
Business Score	Basic	Advanced dimensions + history	Custom score model
AI Advisor	Limited/tips	Contextual insights	Custom advisor
Goals & milestones	Limited	Advanced	Custom
Calendar	Basic dates	Reminders	Workflow reminders
Leaderboard	No/optional	Optional	Private benchmark
Readiness export	No	Yes	Custom reports

9. Information Architecture

Business Hub is integrated into the existing Bisnis Naik Kelas dashboard and must preserve the current dashboard vibe, sidebar structure, card system, dark/light behavior if available, and existing UI components. The Business Hub UI should not feel like a separate application.

Workspace

└ Home



9.1 Dashboard Composition

- Top hero: current level, level name, XP progress, next milestone, mountain/journey visual, and primary CTA.
- Business Score panel: overall score and dimension breakdown.
- Roadmap path: current stage, completed stages, locked future stages, recommended next stage.
- Today Quest: 3-5 actionable checklist items with XP reward.
- Integration status: Website, Accounting, File Manager, SSO, Commerce, AI, Notification.
- Recent activity: completed item, upload, achievement, score change, AI insight.
- Quick actions: upload document, complete checklist, create goal, open accounting, edit website, ask AI.

9.2 UI Tone

The UI should feel encouraging and operational, not childish. Gamification must make business progress motivating without trivializing serious business tasks. The visual language can include level, XP, badges, journey path, progress cards, and achievements, while retaining the professional dashboard sidebar and component system.

10. Core Modules

Module	Purpose	Notes
Home Dashboard	Aggregated gamified command center for the tenant.	Read roadmap, score, quests, integrations, activities, recommendations.
Business Profile	Single business profile metadata for Business Hub.	May mirror or reference profile data from SSO/Website Service.
Roadmap	Step-based business maturity journey.	Stages, levels, requirements, unlock rules.
Checklist	Concrete task system linked to roadmap and service facts.	Manual/automatic completion, evidence, XP reward.

Business Vault	Business document metadata and relationship layer.	Actual files stay in File Manager.
SOP Library	Reusable SOP metadata and clone workflow.	Clone creates metadata/file copy through File Manager when needed.
Template Center	Business templates for legal, finance, marketing, HR, operations.	Browse, preview, clone, link to checklist.
Business Score	Weighted readiness score engine.	Financial, digital, legal, operation, HR, sales, customer.
Gamification Engine	XP, levels, streaks, quests, achievements.	Driven by events and checklist progress.
Goals & Milestones	User goals tied to score/checklist/accounting facts.	Progress can be manual or integration-based.
Business Calendar	Compliance and operational reminders.	Uses Notification service for delivery.
AI Advisor	Contextual recommendation orchestration.	Calls AI Gateway; stores insight metadata only.
Analytics	Product analytics events and internal metrics.	No sensitive payload leakage.

11. Roadmap and Level Model

Business Hub has two related but different concepts: Business Level and Roadmap Stage. Level is gamified progress driven by XP/score. Roadmap Stage is a business maturity phase with explicit requirements.

Stage	Name	Definition
Stage 1	Start	Business identity exists and basic profile is complete.
Stage 2	Organize	Legal, finance, and operational basics are documented.
Stage 3	Grow	Website, customer tracking, and cashflow visibility are active.
Stage 4	Scale Up	SOP, reporting, team role, and repeatable process exist.
Stage 5	Established	Governance, KPI, advanced reporting, and readiness documents exist.
Stage 6	Corporate Ready	Compliance, funding readiness, and management structure are mature.
Stage 7	Enterprise Ready	Future: advanced auditability, integrations, branches, and controls.

11.1 Default Roadmap Categories

- Identity and Digital Presence
- Legal and Compliance
- Financial Management
- Operations and SOP
- Customer and Sales
- People and Organization
- Growth and Readiness

11.2 Roadmap Requirement Types

- `manual_checklist`: user manually completes and may attach evidence.
- `file_evidence`: completion requires a File Manager file reference.
- `website_status`: completion comes from Website Service status.
- `accounting_fact`: completion comes from SME Accounting metric/fact.
- `profile_field`: completion comes from Business Profile fields.
- `ai_review`: AI suggests but user confirms.
- `admin_verified`: platform/admin confirms completion.

12. Checklist Specification

12.1 Checklist Item Fields

Field	Description
<code>id</code>	UUID
<code>organization_id</code>	UUID from SSO context
<code>tenant_id</code>	UUID from tenant context
<code>roadmap_stage_id</code>	UUID
<code>category</code>	identity/legal/finance/operation/customer/hr/growth
<code>title</code>	Short user-facing task title
<code>description</code>	Detailed guidance
<code>completion_type</code>	manual/file_evidence/website_status/accounting_fact/profile_field/ai_review/admin_verified
<code>xp_reward</code>	Integer
<code>priority</code>	low/medium/high/critical
<code>required_for_stage</code>	Boolean
<code>evidence_required</code>	Boolean
<code>evidence_file_id</code>	File Manager ID optional
<code>auto_rule_key</code>	Machine-readable integration rule
<code>status</code>	not_started/in_progress/completed/blocked/skipped
<code>completed_at</code>	UTC timestamp
<code>completed_by</code>	SSO user ID

12.2 Default MVP Checklist Seeds

Category	Checklist	Completion Source	Reward
Identity	Complete business profile	<code>profile_field</code>	+50 XP
Identity	Publish business website	<code>website_status</code>	+300 XP
Identity	Connect custom domain	<code>website_status</code>	+250 XP
Legal	Upload NIB	<code>file_evidence</code>	+150 XP
Legal	Upload NPWP	<code>file_evidence</code>	+150 XP
Finance	Separate business cash/bank account	<code>manual/file_evidence</code>	+150 XP
Finance	Record first income	<code>accounting_fact</code>	+100 XP
Finance	Record first expense	<code>accounting_fact</code>	+100 XP
Finance	Generate first cashflow summary	<code>accounting_fact</code>	+200 XP
Operation	Create SOP Penjualan	<code>template_clone/manual</code>	+200 XP
Operation	Create SOP Pembelian	<code>template_clone/manual</code>	+200 XP

Customer	Create first customer record	future/customer_fact	+100 XP
Growth	Set monthly revenue goal	goal	+100 XP
Growth	Complete monthly business review	manual/ai_review	+150 XP

13. Gamification Engine

Gamification must reward meaningful business progress, not empty clicks. XP-producing actions must be idempotent per tenant/action/resource to avoid farming. Manual completion that gives XP should be auditable and may require evidence for high-value tasks.

13.1 Level Names

Level	Name	XP Threshold	Meaning
1	Starter	0	Business is starting to structure itself.
2	Growing	1,000	Basic profile and first records exist.
3	Organized	3,000	Core documents and routines are forming.
4	Scale Up	7,000	Business has visible systems and repeatable processes.
5	Established	15,000	Management, reporting, and documentation are mature.
6	Corporate Ready	30,000	Business is ready for formal partnership/funding discussion.
7	Enterprise Ready	60,000	Future advanced stage for larger organizations.

13.2 XP Transaction Rules

- Each XP transaction must have source_type, source_id, event_key, idempotency_key, and awarded_at.
- The same event for the same resource must not award XP twice.
- Manual checklist completion can award XP once; reopening and completing again must not duplicate XP unless admin reset is explicit.
- XP can be reversed only through compensating transaction, not by deleting history.
- XP source must be tenant-scoped and auditable.
- XP is not a financial or contractual benefit and must not be convertible to money.

13.3 Quest Types

Quest Type	Description	Reward Pattern
------------	-------------	----------------

Daily Quest	Small action refreshed daily: upload one document, review cashflow, complete profile field.	Low XP, habit forming
Weekly Challenge	Score improvement or checklist cluster completion.	Medium XP
Monthly Mission	Larger readiness objective such as Legal Ready or Finance Ready.	Badge + high XP
Milestone Quest	Roadmap-stage unlock task.	Stage progress
Integration Quest	Encourage connecting/using Website, Accounting, Vault, AI.	Cross-service adoption

13.4 Achievements

- First Website Published
- Legal Ready
- Cashflow Hero
- Document Master
- SOP Starter
- Digital Ready
- Finance Ready
- Operationally Organized
- Scale Up Candidate
- Investor Ready
- Consistent Reviewer
- Premium Power User

14. Business Score Engine

Business Score is a weighted readiness score. It should be explainable. Users must see why a score changed and which next actions improve it. Score must support recalculation after checklist events, integration events, and scheduled jobs.

Dimension	Default Weight	Example Inputs
Digital Presence	15%	Website status, domain, SSL, profile completeness, basic SEO.
Financial Management	20%	Accounting setup, cashflow records, budget, reports, account separation.
Legal & Compliance	15%	NIB, NPWP, business documents, expiry metadata, policies.
Operations	20%	SOP, templates, order/customer process, recurring review.
Customer & Sales	10%	Customer record readiness, repeat order process, future CRM integration.
People & Organization	10%	Team, roles, access control, job description, KPI future.
Growth Readiness	10%	Goals, readiness report, monthly review, AI recommendations resolved.

14.1 Score Formula

```
dimension_score = sum(completed_rule_points) / sum(max_rule_points) * 100
overall_score = sum(dimension_score * dimension_weight)
level_candidate = mapping(overall_score, required_stage_completion, xp_threshold)
```

14.2 Score Rules

- Weights must be configurable via seeded database/config, not hard-coded deep in business logic.
- Each score rule must include key, dimension, points, data source, calculation type, and explanation.
- Score recalculation must be idempotent and tenant-scoped.
- Score history must store previous_score, new_score, changed_dimensions, trigger, and calculation_version.
- Premium can access score history and advanced explanations; Free can access basic score and next actions.

15. Business Vault

Business Vault is not file storage. It is the relationship, category, lifecycle, and checklist linkage layer above Arna File Manager.

15.1 Document Categories

- Legal
- Finance
- Tax
- Branding
- Marketing
- Operations
- SOP
- HR
- Vendor
- Contract
- Product
- Customer
- Compliance
- Funding Readiness
- Other

15.2 BusinessDocumentMetadata Fields

Field	Description
-------	-------------

id	UUID
organization_id	UUID
tenant_id	UUID
file_id	File Manager ID
title	User-facing document name
category	Document category
document_type	nib/npwp/akta/sop/contract/template/report/etc
tags	Array
status	draft/active/archived/expired/replaced
issue_date	Optional date
expiry_date	Optional date
linked_checklist_item_id	Optional UUID
linked_roadmap_stage_id	Optional UUID
visibility	owner/admin/team/custom
created_by	SSO user ID
created_at	UTC
updated_at	UTC

15.3 Vault Rules

- Upload and download must be proxied or delegated to File Manager according to existing platform pattern.
- Business Hub stores file_id and metadata only.
- Business Hub must not store raw file bytes, object storage paths, or permanent signed URLs.
- Deleting metadata must not necessarily delete the physical file unless user explicitly chooses delete-file flow and permission allows.
- Document expiry can create Business Calendar events and Notification reminders.
- Documents can act as evidence for checklist completion.

16. SOP Library and Template Center

SOP Library and Template Center provide guided business assets. They should not become complex editors in MVP. MVP supports browse, preview metadata, clone/copy reference, attach to Vault, and link to checklist.

Asset Type	Examples	MVP Behavior
SOP	Sales SOP, Purchasing SOP, Customer Service SOP, Packing SOP, Recruitment SOP	Browse, preview, clone to Vault metadata/file copy
Template	NDA, Proposal, Job Description, KPI, Budget, Business Plan, Investor Deck	Browse, preview, clone to Vault metadata/file copy
Guide	How to make NIB, how to separate account, monthly review guide	Read-only guidance, checklist links

16.1 Template Metadata Fields

- id, title, description, category, target_stage, target_industry, required_plan, file_template_id, preview_image_file_id, tags, created_at, updated_at, is_active

17. Goals, Milestones, and Business Calendar

17.1 Goals

Goals let users define business outcomes such as monthly revenue target, legal completion, website launch, SOP completion, or readiness score target. Some goals are manual. Others can be linked to Accounting facts, Website status, or checklist completion.

Goal Type	Example	Progress Source
Revenue	Reach IDR 100 million monthly revenue	Accounting summary
Digital	Publish website and connect domain	Website Service
Legal	Complete legal documents	Vault/checklist
Operational	Create 3 core SOPs	SOP/checklist
Readiness	Reach Business Score 80	Score engine
Habit	Complete 4 monthly reviews	Manual/checklist

17.2 Calendar

- Domain expiry from Website Service if exposed.
- Document expiry from Vault metadata.
- Monthly review schedule.
- Tax/compliance reminder templates.
- Contract/vendor expiry.
- Goal milestone target date.
- Notification Service owns delivery; Business Hub owns reminder intent metadata.

18. AI Business Advisor

AI Advisor is an orchestrated advisory feature. Business Hub builds a sanitized context packet and sends it to AI Gateway. AI Gateway performs inference. Business Hub stores insight metadata, recommendation text, related entities, status, and feedback.

18.1 AI Context Sources

- Business Profile fields and industry.
- Roadmap stage and checklist state.
- Business Score dimensions and recent changes.

- Website status: published, domain, SSL, pages, SEO score if available.
- Accounting summary: revenue, expense, cashflow, budget status, trend labels; avoid raw sensitive transaction lines by default.
- Vault document metadata: categories, missing docs, expiry dates; never send raw document contents unless explicitly supported and consented.
- Goals and milestones.
- Recent achievements and blocked tasks.

18.2 Recommendation Lifecycle

generated -> viewed -> accepted -> snoozed -> completed -> dismissed

18.3 AI Safety and Product Rules

- AI must be advisory, not authoritative legal/tax/financial certification.
- Recommendations must include rationale and linked actions.
- Prompt and context versions must be stored for debugging.
- Sensitive payload must be minimized.
- Users must be able to dismiss irrelevant recommendations.
- AI errors must not block dashboard loading.

19. Integrations

Service	Business Hub Uses	Must Not Do
Arna SSO	JWT validation, user identity, organization, roles, permissions	Create its own user/role system
Arna Commerce	Entitlements, quota, plan access, upgrade triggers	Infer access from package name only
Arna File Manager	File upload/download, template/SOP files, vault evidence	Store raw file bytes or direct object storage credentials
Website Service	Website published status, domain, SSL, SEO/basic website facts	Manage website builder or domain directly
SME Accounting	Financial summaries, cashflow facts, budget status, score inputs	Create accounting transactions or read accounting DB
Notification Service	Reminder delivery for calendar/expiry/quest	Send email/WA/push directly
AI Gateway	Insight generation, recommendation generation	Call model provider directly from Business Hub
Pulsar	Async events across services	Require for every MVP request if simple polling/API is enough

19.1 Integration Status Model

- connected: service reachable and tenant context exists.
- warning: service reachable but setup incomplete or quota near limit.
- disconnected: service unavailable or tenant not configured.
- unknown: status cannot be determined; dashboard should degrade gracefully.

20. API Requirements

Use HTTP/JSON API. Base path: /api/v1/business-hub. All protected endpoints require Authorization: Bearer <sso-access-token> and verified organization/tenant context. Response format follows platform standard.

```
{
  "data": {},
  "meta": {},
  "request_id": "uuid"
}

{
  "error": "machine_readable_code",
  "detail": "Human-readable explanation.",
  "request_id": "trace-or-request-id"
}
```

20.1 Core Endpoints

Method	Endpoint	Purpose
GET	/health/live	Process liveness
GET	/health/ready	Readiness check
GET	/api/v1/business-hub/overview	Gamified dashboard aggregate
GET	/api/v1/business-hub/profile	Get business profile
PATCH	/api/v1/business-hub/profile	Update Business Hub profile metadata
GET	/api/v1/business-hub/roadmap	Get roadmap and stages
GET	/api/v1/business-hub/checklists	List checklist items
POST	/api/v1/business-hub/checklists/{id}/start	Mark in progress
POST	/api/v1/business-hub/checklists/{id}/complete	Complete item
POST	/api/v1/business-hub/checklists/{id}/reopen	Reopen item
GET	/api/v1/business-hub/vault/documents	List vault metadata
POST	/api/v1/business-hub/vault/documents	Create metadata after File Manager upload
GET	/api/v1/business-hub/vault/documents/{id}	Document metadata detail
PATCH	/api/v1/business-hub/vault/documents/{id}	Update metadata
DELETE	/api/v1/business-hub/vault/documents/{id}	Archive/delete metadata
GET	/api/v1/business-hub/sop-library	List SOP templates
POST	/api/v1/business-hub/sop-library/{id}/clone	Clone SOP to Vault
GET	/api/v1/business-hub/templates	List templates
POST	/api/v1/business-hub/templates/{id}/clone	Clone template to Vault
GET	/api/v1/business-hub/score	Current score
GET	/api/v1/business-hub/score/history	Score history Premium
POST	/api/v1/business-hub/score/recalculate	Trigger recalculation admin/system

GET	/api/v1/business-hub/quests	Daily/weekly/monthly quests
POST	/api/v1/business-hub/quests/{id}/claim	Claim completed quest reward
GET	/api/v1/business-hub/achievements	Achievements and badges
GET	/api/v1/business-hub/goals	List goals
POST	/api/v1/business-hub/goals	Create goal
PATCH	/api/v1/business-hub/goals/{id}	Update goal
GET	/api/v1/business-hub/calendar/events	List calendar events
POST	/api/v1/business-hub/calendar/events	Create event/reminder
GET	/api/v1/business-hub/ai/insights	List AI insights
POST	/api/v1/business-hub/ai/insights/generate	Generate insight
POST	/api/v1/business-hub/ai/insights/{id}/feedback	Submit feedback
GET	/api/v1/business-hub/integrations/status	Aggregated service status
GET	/api/v1/business-hub/settings	Tenant settings
PATCH	/api/v1/business-hub/settings	Update settings

21. Example API Payloads

21.1 Complete Checklist Item

```
{
  "completion_note": "NIB uploaded and verified by owner",
  "evidence_file_id": "file-manager-id",
  "idempotency_key": "uuid"
}
```

21.2 Create Vault Metadata

```
{
  "file_id": "file-manager-id",
  "title": "NIB PT Example",
  "category": "legal",
  "document_type": "nib",
  "tags": ["legal", "company"],
  "issue_date": "2026-07-01",
  "expiry_date": null,
  "linked_checklist_item_id": "uuid"
}
```

21.3 Create Goal

```
{
  "title": "Reach IDR 100 million monthly revenue",
  "goal_type": "revenue",
  "target_value": "100000000.00",
  "current_value_source": "accounting.monthly_revenue",
}
```

```
"target_date": "2026-12-31"
}
```

22. Event Requirements

Pulsar is recommended for cross-service completion and score updates. API polling can be used in MVP where event producer is unavailable, but the domain model must be event-ready.

22.1 Events Produced

- business_hub.checklist.completed.v1
- business_hub.roadmap.stage_unlocked.v1
- business_hub.xp.awarded.v1
- business_hub.achievement.unlocked.v1
- business_hub.score.updated.v1
- business_hub.document.created.v1
- business_hub.goal.completed.v1
- business_hub.ai_insight.generated.v1
- business_hub.calendar.reminder_due.v1

22.2 Events Consumed

- website.site.published.v1
- website.domain.connected.v1
- website.ssl.active.v1
- accounting.transaction.created.v1
- accounting.cashflow.summary_ready.v1
- accounting.budget.exceeded.v1
- file_manager.file.created.v1
- commerce.subscription.updated.v1
- sso.organization_member.updated.v1

22.3 Standard Event Envelope

```
{
  "event_id": "uuid",
  "event_type": "business_hub.checklist.completed.v1",
  "occurred_at": "2026-07-06T00:00:00Z",
  "producer": "arna-business-hub",
  "organization_id": "uuid",
  "tenant_id": "uuid",
  "correlation_id": "uuid",
  "data": {}
}
```

23. Permissions

Permission	Description
business_hub.dashboard.view	View Business Hub overview
business_hub.profile.view	View business profile
business_hub.profile.manage	Update profile
business_hub.roadmap.view	View roadmap
business_hub.checklist.view	View checklist
business_hub.checklist.manage	Start/complete/reopen checklist
business_hub.vault.view	View document metadata
business_hub.vault.manage	Create/update/archive document metadata
business_hub.sop.view	Browse SOP library
business_hub.sop.clone	Clone SOP to Vault
business_hub.template.view	Browse templates
business_hub.template.clone	Clone templates
business_hub.score.view	View score
business_hub.score.admin_recalculate	Trigger score recalculation
business_hub.goal.view	View goals
business_hub.goal.manage	Manage goals
business_hub.calendar.view	View calendar
business_hub.calendar.manage	Manage reminders
business_hub.ai.view	View AI insights
business_hub.ai.generate	Generate AI insights
business_hub.settings.manage	Update settings

24. Data Model

- BusinessHubTenantSetting
- BusinessProfile
- Roadmap
- RoadmapStage
- RoadmapRequirement
- ChecklistItem
- ChecklistCompletion
- BusinessDocumentMetadata
- SopTemplate
- BusinessTemplate
- BusinessScore
- BusinessScoreDimension
- BusinessScoreRule
- BusinessScoreHistory
- XPTransaction
- LevelState
- Quest
- QuestAssignment
- Achievement
- AchievementUnlock
- Goal

- Milestone
- CalendarEvent
- AllInsight
- IntegrationStatusSnapshot
- ActivityLog
- AuditLog

24.1 Key Model Requirements

- Every tenant-owned table must include `organization_id` and `tenant_id`.
- Unique constraints must include tenant ownership fields where tenant-specific.
- Do not scope queries by `user_id` alone.
- Timestamps must be stored in UTC.
- Soft delete/archive preferred for user-visible records.
- Audit sensitive write operations.
- Use JSONB only for flexible metadata, not as replacement for core relational fields.
- Do not read or write other services databases directly.

24.2 Important Tables

Table	Key Fields
BusinessProfile	id, organization_id, tenant_id, business_name, industry, business_stage, employee_count, monthly_revenue_range, address, logo_file_id, profile_completion_percent
ChecklistItem	id, organization_id, tenant_id, roadmap_stage_id, category, title, completion_type, xp_reward, status, completed_at, evidence_file_id
BusinessDocumentMetadata	id, organization_id, tenant_id, file_id, title, category, document_type, issue_date, expiry_date, linked_checklist_item_id, status
BusinessScore	id, organization_id, tenant_id, overall_score, level, calculation_version, calculated_at
XPTransaction	id, organization_id, tenant_id, source_type, source_id, event_key, points, idempotency_key, awarded_at
AllInsight	id, organization_id, tenant_id, insight_type, title, body, related_entity_type, related_entity_id, status, prompt_version, context_version

25. Background Jobs

- `score_recalculation_job`: recalculates score after checklist/integration changes.
- `quest_assignment_job`: creates daily/weekly quests per tenant.
- `document_expiry_scan_job`: creates reminder events for expiring documents.
- `calendar_reminder_dispatch_job`: sends due reminders to Notification Service.
- `integration_status_snapshot_job`: refreshes service status for dashboard.
- `ai_insight_generation_job`: generates scheduled Premium insights within quota.

- `streak_update_job`: updates review/action streaks.
- `readiness_export_job`: generates Premium readiness report when requested.

26. Frontend Requirements

26.1 Existing Dashboard Integration

- Use existing dashboard shell, sidebar, topbar, auth state, tenant selector, UI cards, modals, buttons, tables, tabs, empty states, and toast system.
- Business Hub must not introduce a separate login or standalone public app.
- Sidebar must remain consistent; Business Hub may be a group or top-level workspace depending on current app convention.
- Use responsive layouts: desktop dashboard first, tablet usable, mobile readable for owner quick actions.
- Dark/dashboard vibe should remain if the current dashboard is dark; gamification accents should be layered through cards and progress components.

26.2 Screens

Screen	Required Content
Overview	Hero level, XP, score, roadmap, quests, integrations, recent activities, AI recommendations
Roadmap	Stage list/path, requirements, locked/unlocked states, stage detail
Checklist	Filterable checklist, category tabs, completion modal, evidence upload link
Documents/Vault	Document grid/table, category filters, metadata form, expiry indicators
SOP Library	Template cards, category filters, preview, clone CTA
Templates	Template cards, preview, clone CTA
Goals	Goal list, create/edit form, progress cards
Calendar	Upcoming reminders, document expiry, monthly review, add reminder
AI Advisor	Insight feed, generate CTA, related action buttons, feedback
Achievements	Badges, unlocked/locked achievements, XP history
Settings	Business Hub preferences, notification preferences, score recalculation admin if allowed

26.3 Empty States

- No roadmap progress: Start by completing your business profile.
- No documents: Upload your first business document to build your Business Vault.
- No accounting connection: Open Accounting to start tracking money and improve Financial Score.
- No website published: Publish your website to improve Digital Presence Score.

- No AI insight: Generate your first recommendation after completing profile and at least one checklist item.
- Free limit reached: user can still view existing data; creation is blocked only for gated feature.

27. Analytics Events

- business_hub_opened
- business_profile_completed
- roadmap_stage_viewed
- checklist_item_started
- checklist_item_completed
- vault_document_created
- sop_template_cloned
- business_template_cloned
- score_viewed
- score_recalculated
- quest_completed
- achievement_unlocked
- goal_created
- calendar_reminder_created
- ai_insight_generated
- ai_insight_accepted
- upgrade_prompt_shown
- upgrade_clicked
- free_limit_reached

Analytics events must include organization_id, tenant_id, user_id, plan, feature, timestamp, and request_id. Do not include sensitive document content, AI prompt raw context, or full financial payloads.

28. Security Requirements

30. Validate JWT signature, issuer, audience, expiry, and required claims.
31. Verify organization membership and tenant access on every protected request.
32. Enforce permission server-side for every action.
33. Check Commerce entitlement for package-controlled features.
34. Use File Manager for file access and respect its permission model.
35. Never log raw JWT, passwords, secrets, raw document contents, or full AI context.
36. Apply rate limit to write endpoints, AI generation, export, and score recalculation.
37. Use request body size limits and strict input validation.
38. Restrict CORS to approved dashboard origins.
39. Keep secrets outside Git and Docker images.

- 40. Audit sensitive writes: checklist completion, document metadata, AI generation, settings, score admin recalculation.
- 41. Provide tenant isolation tests for all tenant-owned resources.

29. Observability

29.1 Health

- GET /health/live confirms process is running.
- GET /health/ready checks database, Redis if required, critical configuration, SSO JWKS/public key availability, and optionally core dependency reachability.

29.2 Structured Logs

- timestamp
- level
- service
- environment
- request_id
- organization_id
- tenant_id
- user_id
- operation
- resource_type
- resource_id
- result
- duration_ms
- error_code

29.3 Metrics

- http_requests_total
- http_request_duration_seconds
- http_errors_total
- business_hub_checklist_completed_total
- business_hub_xp_awarded_total
- business_hub_score_recalculated_total
- business_hub_ai_insights_generated_total
- business_hub_entitlement_denied_total
- business_hub_integration_status_failures_total
- db_query_duration_seconds
- background_job_duration_seconds
- pulsar_consumer_lag

30. Resilience Requirements

- Every external service client must define connect timeout, read timeout, retry behavior, maximum attempts, and backoff.
- Retry only transient failures; never retry indefinitely.
- Integration status failure must degrade dashboard gracefully.
- AI Gateway failure must not block non-AI dashboard data.
- File Manager failure must block only file-dependent actions.
- Score recalculation must be repeatable and idempotent.
- Event consumers must tolerate duplicate events.
- Event publishing failure must not corrupt primary database transaction.
- Use idempotency key for mutating endpoints where duplicate user submissions are likely.

31. Recommended Stack

Layer	Recommendation
Backend	Python, Django, Django REST Framework
Database	PostgreSQL
Cache/Jobs	Redis + Celery or Django-Q depending on existing boilerplate
Events	Pulsar client when async integration is enabled
OpenAPI	drf-spectacular
Testing	pytest, pytest-django, factory_boy
Frontend	Existing Bisnis Naik Kelas dashboard stack and component system
Deployment	Docker image deployed as Docker Swarm service
Reverse proxy	Nginx Proxy Manager/API gateway pattern if public route needed

32. Repository Structure

```
arna_business_hub/
├── README.md
├── docs/
│   ├── ARCHITECTURE.md
│   ├── API.md
│   ├── EVENTS.md
│   ├── SCORE_ENGINE.md
│   └── ADR/
├── Dockerfile
├── Jenkinsfile
├── .env.example
├── source/
│   ├── business_hub/
│   │   ├── profiles/
│   │   ├── roadmap/
│   │   ├── checklist/
│   │   ├── vault/
│   │   └── templates/
```



```

├── gamification/
├── scoring/
├── goals/
├── calendar/
├── ai_advisor/
├── integrations_status/
├── settings/
├── integrations/
│   ├── sso/
│   ├── commerce/
│   ├── file_manager/
│   ├── website/
│   ├── accounting/
│   ├── notification/
│   ├── ai_gateway/
│   └── pulsar/
├── common/
├── config/
├── manage.py
└── tests/

```

33. Environment Variables

```

SERVICE_NAME=arna-business-hub
ENVIRONMENT=development
DEBUG=False
SECRET_KEY=replace-me
ALLOWED_HOSTS=localhost,business-hub.bisnisnaikkelas.com

```

```

POSTGRES_DB=arna_business_hub_db
POSTGRES_USER=postgres
POSTGRES_PASSWORD=replace-me
POSTGRES_HOST=postgres
POSTGRES_PORT=5432
REDIS_URL=redis://redis:6379/0

```

```

SSO_BASE_URL=https://sso.arnatech.id
SSO_JWT_AUDIENCE=arna-business-hub
SSO_JWT_ISSUER=sso.arnatech.id
SSO_PUBLIC_KEY_PATH=/run/secrets/sso_public.pem

```

```

COMMERCE_BASE_URL=https://product.arnatech.id/api/v1
FILE_MANAGER_BASE_URL=https://storage.arnatech.id
WEBSITE_SERVICE_BASE_URL=https://bisnisnaikkelas.com/api/v1
ACCOUNTING_SERVICE_BASE_URL=https://accounting.bisnisnaikkelas.com/api/v1
NOTIFICATION_SERVICE_BASE_URL=https://notification.arnatech.id/api/v1
AI_GATEWAY_BASE_URL=https://ai.arnatech.id/api/v1

```

```

PULSAR_SERVICE_URL=pulsar://broker:6650
PULSAR_ENABLED=False
PULSAR_SUBSCRIPTION=arna-business-hub-consumer

```

```

DEFAULT_CURRENCY=IDR
SCORE_CALCULATION_VERSION=1

```

AI_CONTEXT_MAX_TOKENS=8000
AI_MONTHLY_INSIGHT_DEFAULT_QUOTA=10

34. MVP Scope

34.1 Included

- Backend service setup
- SSO authentication and tenant middleware
- Commerce entitlement client
- File Manager metadata integration
- Website status client
- Accounting summary client
- Business Profile
- Roadmap seed and checklist seed
- Checklist completion with XP
- Basic Business Vault metadata
- Basic SOP/template metadata browse
- Business Score calculation v1
- Gamified dashboard overview
- Quest assignment basics
- Achievement unlock basics
- Goals basic CRUD
- Calendar basic CRUD and reminder intent
- Integration status endpoint
- OpenAPI docs
- Health endpoints
- Dockerfile and Jenkinsfile
- Core unit/integration tests
- Frontend dashboard integration screens

34.2 Excluded from MVP

- Custom Enterprise roadmap builder
- Full document editor
- Complex approval workflow
- Public leaderboard
- Deep benchmarking
- CRM implementation
- Inventory implementation
- HR implementation
- Legal/tax filing submission
- Direct payment integration

- Direct AI model provider integration
- Native mobile app

35. Phase Plan

Phase	Deliverables
Phase 0 - Service Design	Finalize PRD, architecture, OpenAPI draft, event contract, score rule seed, permission matrix.
Phase 1 - Foundation	Repo, Django/DRF, DB, health, Docker, Jenkins, SSO middleware, tenant isolation, Commerce/File Manager clients.
Phase 2 - Core Business Hub	Business profile, roadmap, checklist, vault metadata, seed data, basic dashboard aggregate.
Phase 3 - Gamification and Score	XP, level, achievements, quests, score engine, score history, score recalculation job.
Phase 4 - Integrations	Website status, Accounting facts, File Manager evidence, Notification reminder, integration status.
Phase 5 - AI Advisor	AI context builder, AI Gateway client, insight lifecycle, quota enforcement, feedback.
Phase 6 - Frontend Integration	Overview, roadmap, checklist, vault, templates, goals, calendar, AI, achievements, settings.
Phase 7 - Production Readiness	Security review, tenant isolation tests, observability, deployment, backup/rollback, UAT.

36. Testing Requirements

36.1 Unit Tests

- Business score calculation
- Score rule weighting
- XP idempotency
- Level threshold mapping
- Quest assignment
- Achievement unlock
- Checklist completion state transition
- Vault metadata validation
- Entitlement decision logic
- AI context sanitizer
- Integration status mapping
- Calendar reminder due calculation

36.2 Authorization and Tenant Isolation Tests

- Unauthenticated request is rejected.
- Invalid JWT is rejected.
- Expired JWT is rejected.

- Missing permission is rejected.
- Tenant A cannot read Tenant B checklist.
- Tenant A cannot update Tenant B document metadata.
- Tenant A cannot claim Tenant B quest reward.
- Tenant context header mismatch is rejected.
- Entitlement denied blocks Premium-only action.
- Free user can view existing data after reaching quota but cannot create gated records.

36.3 API Integration Tests

- Get dashboard overview
- Update profile
- List roadmap
- Complete manual checklist
- Complete file-evidence checklist
- Create vault metadata
- Clone SOP template
- Calculate score
- Award XP once only
- Unlock achievement
- Create goal
- Create calendar event
- Generate AI insight with mocked AI Gateway
- Handle Website Service unavailable
- Handle Accounting Service unavailable
- Enforce Free/Premium feature matrix

36.4 Frontend QA Scenarios

- Empty first-login dashboard
- Profile incomplete dashboard
- Free user reaching vault limit
- Premium user generating AI insight
- Website disconnected state
- Accounting disconnected state
- File Manager upload failure
- Checklist evidence required modal
- Quest complete and XP animation/state update
- Achievement unlocked state
- Mobile dashboard layout
- Dark mode/card contrast if supported

37. Acceptance Criteria

- 42. User can open Business Hub from existing Bisnis Naik Kelas dashboard.
- 43. Service authenticates using Arna SSO and verifies tenant context.
- 44. Service checks Arna Commerce entitlement before package-controlled actions.
- 45. User can view gamified overview with level, XP, score, quests, roadmap, integrations, and recent activities.
- 46. User can complete checklist manually and with File Manager evidence.
- 47. Website published/domain status can auto-complete relevant checklist when integration data is available.
- 48. Accounting facts can improve Financial Score when integration data is available.
- 49. Business Vault stores metadata and file_id only.
- 50. SOP/template clone flow creates/link metadata correctly.
- 51. Business Score recalculates and stores history according to package access.
- 52. XP and achievements are awarded idempotently.
- 53. AI Advisor generates contextual recommendations through AI Gateway, not direct model provider calls.
- 54. Calendar reminder intent can be created and dispatched through Notification Service when enabled.
- 55. OpenAPI documentation is available.
- 56. Core tests pass in CI.
- 57. Service runs in Docker and can be deployed as Docker Swarm service.

38. Risks and Mitigations

Risk	Mitigation
Gamification feels childish	Use professional business language, restrained visuals, and meaningful progress rewards.
Users complete checklist without real evidence	Require evidence for high-value legal/finance tasks and store audit trail.
Score feels arbitrary	Expose score explanation, rules, dimensions, and next actions.
Cross-service dependency slows dashboard	Use cached integration snapshots and graceful degradation.
Duplicate XP or farming	Use idempotency keys and source-resource uniqueness.
Tenant data leakage	Enforce tenant middleware, tenant-scoped queries, and cross-tenant tests.
AI gives risky advice	Limit to recommendations, add disclaimers, link to concrete actions, avoid legal/tax guarantee.
Business Hub duplicates other services	Maintain strict service boundary and integration contract.

39. Definition of Done

- 58. Service responsibility and non-responsibility are documented.

59. Authentication uses Arna SSO.
60. Authorization and permissions are enforced server-side.
61. Organization and tenant isolation are enforced.
62. Commerce entitlements are enforced.
63. File attachments use Arna File Manager.
64. Website and Accounting integrations use API/event contracts, not database access.
65. OpenAPI documentation is available and current.
66. Score, XP, checklist, and tenant isolation logic have tests.
67. Frontend screens match existing dashboard component style.
68. Docker image is reproducible.
69. Jenkins pipeline runs tests before build/deploy.
70. Health checks are configured.
71. Structured logs and metrics support troubleshooting.
72. No secrets are committed or baked into image.
73. Security review is completed.
74. Backup, migration, rollback, and seed behavior are documented.

40. Appendix A - Suggested Score Rule Seed

Rule Key	Dimension	Points	Source
digital.website_published	Digital Presence	20	Website Service: site published
digital.domain_connected	Digital Presence	20	Website Service: custom domain active
digital.ssl_active	Digital Presence	15	Website Service: SSL active
legal.nib_uploaded	Legal	25	Vault document type nib exists
legal.npwp_uploaded	Legal	25	Vault document type npwp exists
finance.first_income	Financial	15	Accounting has at least one income transaction
finance.first_expense	Financial	15	Accounting has at least one expense transaction
finance.cashflow_ready	Financial	25	Accounting cashflow summary available
operation.sop_sales	Operations	20	SOP Penjualan exists
operation.sop_purchase	Operations	15	SOP Pembelian exists
growth.goal_created	Growth Readiness	15	At least one active goal
growth.monthly_review	Growth Readiness	20	Monthly review checklist completed

41. Appendix B - Implementation Notes for Engineers

- Prefer service-layer methods for score and XP mutations so API, event consumer, and jobs share the same logic.

- Use database transaction boundaries for checklist completion plus XP award plus activity log.
- Keep seed data versioned and repeatable.
- Treat integration clients as ports/adapters so tests can mock external services easily.
- Make dashboard overview endpoint aggregate from cached/materialized internal state where possible.
- Do not make the frontend responsible for entitlement enforcement; frontend only reflects server decisions.
- Add management commands for seed roadmap, rebuild score, rebuild XP projection, and refresh integration status.
- Use feature flags for AI Advisor and leaderboard.

42. Appendix C - Glossary

Term	Definition
Business Hub	Central orchestration workspace for business progress.
Roadmap	Business maturity path composed of stages.
Checklist	Concrete action items linked to roadmap and score.
Business Vault	Metadata layer for business documents stored in File Manager.
Business Score	Weighted readiness score across business dimensions.
XP	Gamified reward points for meaningful progress.
Quest	Recommended action with reward and time window.
Achievement	Badge unlocked by completing meaningful milestone.
AI Advisor	Contextual recommendation feature powered through AI Gateway.
Tenant	Business/site context under an organization.